

Database Management System

EXPERIMENT 6 SINGLE ROW FUNCTIONS

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Let's verify our data and add some more for better demonstration:

```
-- Check current employees
SELECT COUNT(*) FROM employees;
```

Expected Output:

```
COUNT(*)
-----
        19
```

Exercise Solutions:

1. Display current date

```
SELECT SYSDATE AS "Date" FROM dual;
```

Expected Output:

```
Date
-----
```

23-JAN-24

2. Display employee details with 15.5% salary increase

```
SELECT employee_id, last_name, salary,  
       ROUND(salary * 1.155) AS "New Salary"  
FROM employees;
```

Expected Output:

EMPLOYEE_ID	LAST_NAME	SALARY	New Salary
100	King	24000	27720
101	Kochhar	17000	19635
102	De Haan	17000	19635
103	Hunold	9000	10395
104	Ernst	6000	6930
107	Lorentz	4200	4851
124	Mourgos	5800	6699
141	Rajs	3500	4043
142	Davies	3100	3581
143	Matos	2600	3003
144	Vargas	2500	2888
149	Zlotkey	10500	12128
174	Abel	11000	12705
176	Taylor	8600	9933
200	Whalen	4400	5082
201	Hartstein	13000	15015
202	Fay	6000	6930
205	Higgins	12000	13860

3. Add a column showing the salary increase amount

```

SELECT employee_id, last_name, salary,
       ROUND(salary * 1.155) AS "New Salary",
       ROUND(salary * 1.155) - salary AS "Increase"
FROM employees;

```

Expected Output:

EMPLOYEE_ID	LAST_NAME	SALARY	New Salary	Increase
100	King	24000	27720	3720
101	Kochhar	17000	19635	2635
102	De Haan	17000	19635	2635
103	Hunold	9000	10395	1395
104	Ernst	6000	6930	930
107	Lorentz	4200	4851	651
124	Mourgos	5800	6699	899
141	Rajs	3500	4043	543
142	Davies	3100	3581	481
143	Matos	2600	3003	403
144	Vargas	2500	2888	388

1628	149 Zlotkey	10500	12128
1705	174 Abel	11000	12705
1333	176 Taylor	8600	9933
682	200 Whalen	4400	5082
2015	201 Hartstein	13000	15015
930	202 Fay	6000	6930
1860	205 Higgins	12000	13860

4. Display formatted names and lengths for names starting with J, A, or M

```
SELECT INITCAP(last_name) AS "Last Name",
       LENGTH(last_name) AS "Name Length"
FROM employees
WHERE UPPER(SUBSTR(last_name, 1, 1)) IN ('J', 'A', 'M')
ORDER BY last_name;
```

Expected Output:

Last Name	Name Length
Abel	4
Davies	6
Hartstein	9
Matos	5
Mourgos	7
Rajs	4

5. Prompt user for starting letter and display matching employees

```
-- This will prompt for input
SELECT last_name, LENGTH(last_name) AS "Name Length"
FROM employees
WHERE UPPER(SUBSTR(last_name, 1, 1)) =
UPPER('&start_letter')
ORDER BY last_name;
```

When prompted with 'H': Expected Output:

```
Enter value for start_letter: H
old   3: WHERE UPPER(SUBSTR(last_name, 1, 1)) =
UPPER('&start_letter')
new   3: WHERE UPPER(SUBSTR(last_name, 1, 1)) = UPPER('H')
LAST_NAME                                Name Length
-----
Hartstein                                9
Higgins                                  7
Hunold                                    6
```

6. Calculate months worked for each employee

```
SELECT last_name,
       CEIL(MONTHS_BETWEEN(SYSDATE, hire_date)) AS
MONTHS_WORKED
FROM employees
ORDER BY MONTHS_WORKED;
```

Expected Output:

LAST_NAME	MONTHS_WORKED
King	439
Whalen	436
Kochhar	412
Hunold	408
Ernst	392
De Haan	372
Higgins	355
Rajs	339
Hartstein	334
Abel	331
Fay	317
Davies	323
Matos	310
Taylor	309
Vargas	305
Mourgos	290
Lorentz	298
Zlotkey	287

7. Create "Dream Salaries" report

```
SELECT last_name || ' earns ' || TO_CHAR(salary) ||
       ' monthly but wants ' || TO_CHAR(salary * 3) AS
"Dream Salaries"
FROM employees;
```

Expected Output:

Dream Salaries

King earns 24000 monthly but wants 72000
 Kochhar earns 17000 monthly but wants 51000
 De Haan earns 17000 monthly but wants 51000
 Hunold earns 9000 monthly but wants 27000
 Ernst earns 6000 monthly but wants 18000
 Lorentz earns 4200 monthly but wants 12600
 Mourgos earns 5800 monthly but wants 17400
 Rajs earns 3500 monthly but wants 10500
 Davies earns 3100 monthly but wants 9300
 Matos earns 2600 monthly but wants 7800
 Vargas earns 2500 monthly but wants 7500
 Zlotkey earns 10500 monthly but wants 31500
 Abel earns 11000 monthly but wants 33000
 Taylor earns 8600 monthly but wants 25800
 Whalen earns 4400 monthly but wants 13200
 Hartstein earns 13000 monthly but wants 39000
 Fay earns 6000 monthly but wants 18000
 Higgins earns 12000 monthly but wants 36000

8. Display salary with left-padded \$ symbol

```

SELECT last_name,
       LPAD('$' || TO_CHAR(salary), 15, '$') AS SALARY
FROM employees;
  
```

Expected Output:

LAST_NAME	SALARY
King	\$\$\$\$\$\$\$\$\$24000
Kochhar	\$\$\$\$\$\$\$\$\$17000
De Haan	\$\$\$\$\$\$\$\$\$17000
Hunold	\$\$\$\$\$\$\$\$\$9000

Ernst	\$\$\$\$\$\$\$\$\$6000
Lorentz	\$\$\$\$\$\$\$\$\$4200
Mourgos	\$\$\$\$\$\$\$\$\$5800
Rajs	\$\$\$\$\$\$\$\$\$3500
Davies	\$\$\$\$\$\$\$\$\$3100
Matos	\$\$\$\$\$\$\$\$\$2600
Vargas	\$\$\$\$\$\$\$\$\$2500
Zlotkey	\$\$\$\$\$\$\$\$\$10500
Abel	\$\$\$\$\$\$\$\$\$11000
Taylor	\$\$\$\$\$\$\$\$\$8600
Whalen	\$\$\$\$\$\$\$\$\$4400
Hartstein	\$\$\$\$\$\$\$\$\$13000
Fay	\$\$\$\$\$\$\$\$\$6000
Higgins	\$\$\$\$\$\$\$\$\$12000

G. Display hire date and salary review date (first Monday after 6 months)

```

SELECT last_name,
       TO_CHAR(hire_date, 'fmDay, "the" fmDDth "of" Month,
YYYY') AS hire_date,
       TO_CHAR(NEXT_DAY(ADD_MONTHS(hire_date, 6), 'MONDAY'),
       'fmDay, "the" fmDDth "of" Month, YYYY') AS
REVIEW
FROM employees;
```

Expected Output:

LAST_NAME	HIRE_DATE
REVIEW	
King	Tuesday, the 17th of June, 1987
Monday, the 21st of December, 1987	

Kochhar	Thursday, the 21st of September,
1989	Monday, the 26th of March, 1990
De Haan	Wednesday, the 13th of January,
1993	Monday, the 19th of July, 1993
Hunold	Wednesday, the 3rd of January,
1990	Monday, the 9th of July, 1990
Ernst	Tuesday, the 21st of May, 1991
Monday, the 25th of November, 1991	
Lorentz	Sunday, the 7th of February, 1999
Monday, the 9th of August, 1999	
Mourgos	Tuesday, the 16th of November,
1999	Monday, the 22nd of May, 2000
Rajs	Tuesday, the 17th of October, 1995
Monday, the 22nd of April, 1996	
Davies	Wednesday, the 29th of January,
1997	Monday, the 4th of August, 1997
Matos	Sunday, the 15th of March, 1998
Monday, the 21st of September, 1998	
Vargas	Thursday, the 9th of July, 1998
Monday, the 11th of January, 1999	
Zlotkey	Saturday, the 29th of January,
2000	Monday, the 31st of July, 2000
Abel	Saturday, the 11th of May, 1996
Monday, the 11th of November, 1996	
Taylor	Tuesday, the 24th of March, 1998
Monday, the 28th of September, 1998	
Whalen	Thursday, the 17th of September,
1987	Monday, the 21st of March, 1988
Hartstein	Saturday, the 17th of February,
1996	Monday, the 19th of August, 1996
Fay	Sunday, the 17th of August, 1997
Monday, the 16th of February, 1998	
Higgins	Tuesday, the 7th of June, 1994

Monday, the 12th of December, 1994

10. Display day of week for hire date, ordered starting with Monday

```
SELECT last_name, hire_date,  
       TO_CHAR(hire_date, 'fmDay') AS DAY  
FROM employees  
ORDER BY CASE TO_CHAR(hire_date, 'DY',  
    'NLS_DATE_LANGUAGE=ENGLISH')  
          WHEN 'MON' THEN 1  
          WHEN 'TUE' THEN 2  
          WHEN 'WED' THEN 3  
          WHEN 'THU' THEN 4  
          WHEN 'FRI' THEN 5  
          WHEN 'SAT' THEN 6  
          WHEN 'SUN' THEN 7  
END;
```

Expected Output:

LAST_NAME	HIRE_DATE	DAY
Mourgos	16-NOV-99	Tuesday
Taylor	24-MAR-98	Tuesday
Higgins	07-JUN-94	Tuesday
Rajs	17-OCT-95	Tuesday
Ernst	21-MAY-91	Tuesday
Davies	29-JAN-97	Wednesday
De Haan	13-JAN-93	Wednesday
King	17-JUN-87	Wednesday
Hunold	03-JAN-90	Wednesday
Vargas	09-JUL-98	Thursday
Kochhar	21-SEP-89	Thursday

Whalen	17-SEP-87	Thursday
Hartstein	17-FEB-96	Saturday
Zlotkey	29-JAN-00	Saturday
Abel	11-MAY-96	Saturday
Lorentz	07-FEB-99	Sunday
Fay	17-AUG-97	Sunday
Matos	15-MAR-98	Sunday

Additional Single Row Function Examples:

Character Functions:

11. Case conversion functions

```
SELECT last_name,
       LOWER(last_name) AS lower_name,
       UPPER(last_name) AS upper_name,
       INITCAP(last_name) AS initcap_name
FROM employees
WHERE ROWNUM <= 5;
```

Expected Output:

LAST_NAME INITCAP_NAME	LOWER_NAME	UPPER_NAME
King	king	KING
King		
Kochhar	kochhar	KOCHHAR
Kochhar		
De Haan	de haan	DE HAAN
De Haan		
Hunold	hunold	HUNOLD

Hunold		
Ernst	ernst	ERNST
Ernst		

12. String manipulation functions

```
SELECT last_name,
       SUBSTR(last_name, 1, 3) AS first_3_chars,
       LENGTH(last_name) AS name_length,
       INSTR(last_name, 'a') AS position_of_a,
       REPLACE(last_name, 'a', '*') AS replaced_a
FROM employees
WHERE ROWNUM <= 5;
```

Expected Output:

LAST_NAME	FIRST_3_CHARS	NAME_LENGTH
POSITION_OF_A	REPLACED_A	
-----	-----	-----
-----	-----	-----
King	Kin	4
0 King		
Kochhar	Koc	7
5 Kochh*r		
De Haan	De	7
4 De H**n		
Hunold	Hun	6
0 Hunold		
Ernst	Ern	5
0 Ernst		

Numeric Functions:

13. Rounding and truncation functions

```
SELECT salary,
       ROUND(salary, -3) AS rounded_thousands,
       TRUNC(salary, -3) AS truncated_thousands,
       MOD(salary, 1000) AS remainder
FROM employees
WHERE ROWNUM <= 5;
```

Expected Output:

SALARY	ROUNDED_THOUSANDS	TRUNCATED_THOUSANDS	REMAINDER
24000	24000	24000	0
17000	17000	17000	0
17000	17000	17000	0
9000	9000	9000	0
6000	6000	6000	0

Date Functions:

14. Date arithmetic and functions

```
SELECT last_name, hire_date,
       ADD_MONTHS(hire_date, 6) AS after_6_months,
       LAST_DAY(hire_date) AS month_end,
       NEXT_DAY(hire_date, 'FRIDAY') AS next_friday
FROM employees
WHERE ROWNUM <= 5;
```

Expected Output:

LAST_NAME	HIRE_DATE	AFTER_6_MO	MONTH_END
NEXT_FRIDA			

King	17-JUN-87	17-DEC-87	30-JUN-87
19-JUN-87			
Kochhar	21-SEP-89	21-MAR-90	30-SEP-89
22-SEP-89			
De Haan	13-JAN-93	13-JUL-93	31-JAN-93
15-JAN-93			
Hunold	03-JAN-90	03-JUL-90	31-JAN-90
05-JAN-90			
Ernst	21-MAY-91	21-NOV-91	31-MAY-91
24-MAY-91			

Conversion Functions:

15. TO_CHAR with dates

```
SELECT last_name, hire_date,
       TO_CHAR(hire_date, 'DD-MON-YYYY') AS simple_format,
       TO_CHAR(hire_date, 'Day, Month DD, YYYY') AS
long_format,
       TO_CHAR(hire_date, 'YYYY"Q"Q') AS quarter
FROM employees
WHERE ROWNUM <= 5;
```

Expected Output:

LAST_NAME	HIRE_DATE	SIMPLE_FORMAT	QUARTER
LONG_FORMAT			
King	17-JUN-87	17-JUN-1987	Wednesday,
June 17, 1987	1987Q2		
Kochhar	21-SEP-89	21-SEP-1989	Thursday,

September 21, 1989	1989Q3		
De Haan	13-JAN-93	13-JAN-1993	Wednesday,
January 13, 1993	1993Q1		
Hunold	03-JAN-90	03-JAN-1990	Wednesday,
January 03, 1990	1990Q1		
Ernst	21-MAY-91	21-MAY-1991	Tuesday,
May 21, 1991	1991Q2		

16. TO_CHAR with numbers

```
SELECT last_name, salary,
       TO_CHAR(salary, '$99,999.00') AS currency_format,
       TO_CHAR(salary, '099999') AS leading_zeros,
       TO_CHAR(salary, '9.999EEEE') AS scientific
FROM employees
WHERE ROWNUM <= 5;
```

Expected Output:

LAST_NAME	SALARY	CURRENCY_FORMAT	LEADING_ZEROS	SCIENTIFIC
King	24000	\$24,000.00	024000	2.400E+04
Kochhar	17000	\$17,000.00	017000	1.700E+04
De Haan	17000	\$17,000.00	017000	1.700E+04
Hunold	9000	\$9,000.00	009000	9.000E+03
Ernst	6000	\$6,000.00	006000	

6.000E+03

17. NVL and NVL2 functions

```
SELECT last_name, salary, commission_pct,  
       NVL(commission_pct, 0) AS nvl_commission,  
       NVL2(commission_pct, 'Has Commission', 'No  
Commission') AS commission_status  
FROM employees  
WHERE ROWNUM <= 8;
```

Expected Output:

LAST_NAME	SALARY	COMMISSION_PCT	NVL_COMMISSION	COMMISSION_STATUS
King	24000	0	No	
Commission				
Kochhar	17000	0	No	
Commission				
De Haan	17000	0	No	
Commission				
Hunold	9000	0	No	
Commission				
Ernst	6000	0	No	
Commission				
Lorentz	4200	0	No	
Commission				
Mourgos	5800	0	No	
Commission				
Rajs	3500	0	No	

Commission

Comprehensive Example:

18. Employee information with formatted output

```
SELECT
    INITCAP(last_name) || ', ' || INITCAP(first_name) AS
"Employee Name",
    'Department: ' || department_id AS "Department",
    TO_CHAR(salary, '$99,999') AS "Monthly Salary",
    TO_CHAR(salary * 12, '$999,999') AS "Annual Salary",
    TO_CHAR(hire_date, 'MM/DD/YYYY') AS "Hire Date",
    ROUND(MONTHS_BETWEEN(SYSDATE, hire_date)/12, 1) || '
years' AS "Service"
FROM employees
ORDER BY last_name;
```

Expected Output:

Employee Name	Department	Monthly
Salary Annual Salary Hire Date Service		
Abel, Ellen	Department: 80	\$11,000
\$132,000 05/11/1996 27.6 years		
Davies, Curtis	Department: 50	\$3,100
\$37,200 01/29/1997 26.9 years		
De Haan, Lex	Department: 90	\$17,000
\$204,000 01/13/1993 31 years		
Ernst, Bruce	Department: 60	\$6,000
\$72,000 05/21/1991 32.6 years		
Fay, Pat	Department: 20	\$6,000

\$72,000	08/17/1997	26.4 years	
Hartstein, Michael	Department: 20		\$13,000
\$156,000	02/17/1996	27.9 years	
Higgins, Shelley	Department: 110		\$12,000
\$144,000	06/07/1994	29.6 years	
Hunold, Alexander	Department: 60		\$9,000
\$108,000	01/03/1990	34 years	
King, Steven	Department: 90		\$24,000
\$288,000	06/17/1987	36.6 years	
Kochhar, Neena	Department: 90		\$17,000
\$204,000	09/21/1989	34.3 years	
Lorentz, Diana	Department: 60		\$4,200
\$50,400	02/07/1999	24.9 years	
Matos, Randall	Department: 50		\$2,600
\$31,200	03/15/1998	25.8 years	
Mourgos, Kevin	Department: 50		\$5,800
\$69,600	11/16/1999	24.2 years	
Rajs, Trenna	Department: 50		\$3,500
\$42,000	10/17/1995	28.2 years	
Taylor, Jonathan	Department: 80		\$8,600
\$103,200	03/24/1998	25.8 years	
Vargas, Peter	Department: 50		\$2,500
\$30,000	07/09/1998	25.5 years	
Whalen, Jennifer	Department: 10		\$4,400
\$52,800	09/17/1987	36.3 years	
Zlotkey, Eleni	Department: 80		\$10,500
\$126,000	01/29/2000	23.9 years	

Summary of Single Row Functions:

1. **Character Functions** - UPPER, LOWER, INITCAP, SUBSTR, LENGTH, INSTR, CONCAT, LPAD, RPAD, TRIM, REPLACE
2. **Number Functions** - ROUND, TRUNC, MOD, CEIL, FLOOR

3. **Date Functions** - SYSDATE, MONTHS_BETWEEN, ADD_MONTHS, NEXT_DAY, LAST_DAY, ROUND, TRUNC
4. **Conversion Functions** - TO_CHAR, TO_NUMBER, TO_DATE
5. **General Functions** - NVL, NVL2, COALESCE, CASE, DECODE